Data Analytics Project: Railway Gauge Trends

Dataset Used:railway_gauges.csv

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This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)
Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.**Modules Used**

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (□ Easy)

Description:
we work with a CSV file that contains railway gauge data such as Broad Gauge, Metre Gauge, Narrow Gauge, and Total track length over different years. Before analyzing or plotting graphs, the data must be properly loaded and cleaned. This ensures accuracy and avoids errors during analysis.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

Result of scenario 1:

```
top 5 Rows are
       Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
0  1964-65         3521         2891           464   6876
1  1965-66         3603         2913           470   6986
2  1966-67         3615         2911           481   7007
3  1967-68         3601         2920           506   7027
4  1968-69         3603         2921           508   7032

missing values are
```

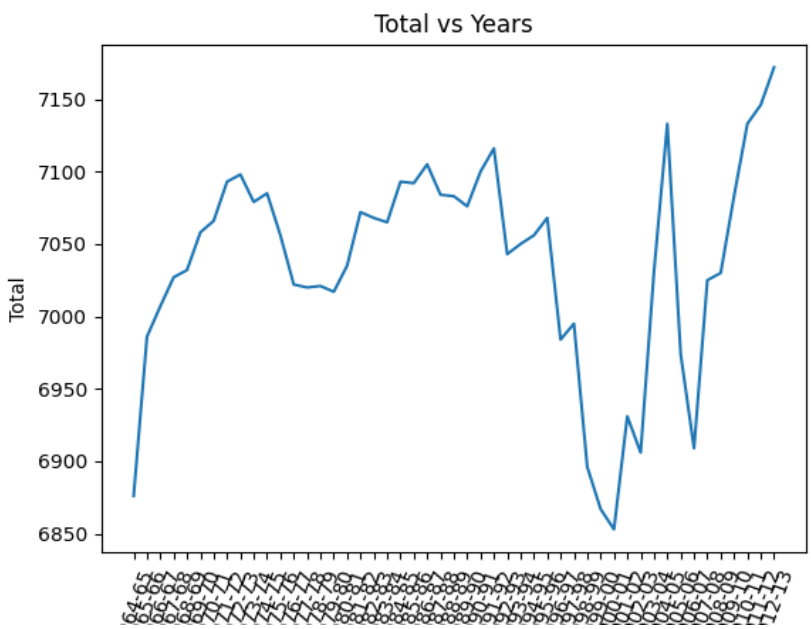
Conclusion:
Basic Data Loading and Cleaning is an essential first step in working with railway gauge data. It involves importing the CSV file, understanding its structure, handling missing values, and converting data into the correct format. These steps ensure that the dataset becomes accurate, consistent, and reliable.
After proper cleaning, the data is ready for analysis and visualization, leading to correct insights about railway gauges.

Scenario 2: Simple Visualization (□ Easy)

Description:
Simple visualization means representing data using basic graphs so that it is easy to understand. In railway gauge data, we use charts to show how track lengths change over years.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing **Total** tracks over the **Years**.
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Graph:



Conclusion:

Simple visualization helps in presenting railway gauge data in an easy and clear graphical form. It makes it simple to understand trends, such as growth or decline over the years, without analyzing complex tables.

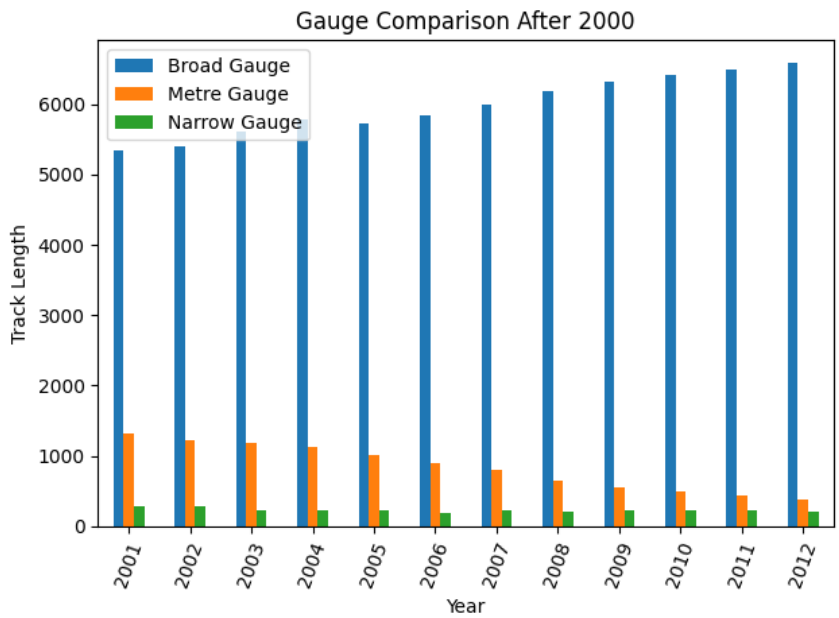
Overall, it improves understanding, saves time, and helps in making better data analysis decisions**.**

Scenario 3: Filtering + Bar Chart (□ Medium)

Description:
In this task, we focus on analyzing specific railway data by selecting only required values (filtering) and then representing it using a bar chart for easy comparison.

Task	Description
1.	Filter the dataset to include only years *after* 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a **grouped bar chart** comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph:



Conclusion:

Filtering allows us to focus on specific and relevant railway data, while a bar chart helps compare different gauges clearly.
Together, they make it easy to identify trends and see which railway gauge has the highest contribution in a simple and visual way**.**

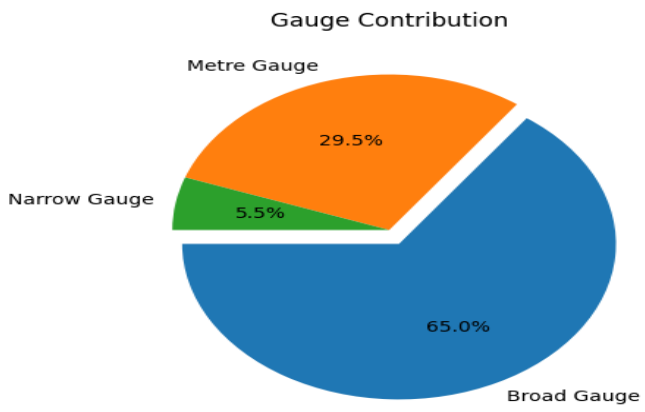
Scenario 4: Feature Engineering + Pie Chart (□ Medium)

Description:

we calculate the total of each railway gauge across all years.These totals help us understand the overall contribution of each gauge type.A pie chart is used to represent these contributions as percentages.From the chart, we can easily identify which gauge dominates the railway network.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a **pie chart** using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph:



Conclusion:

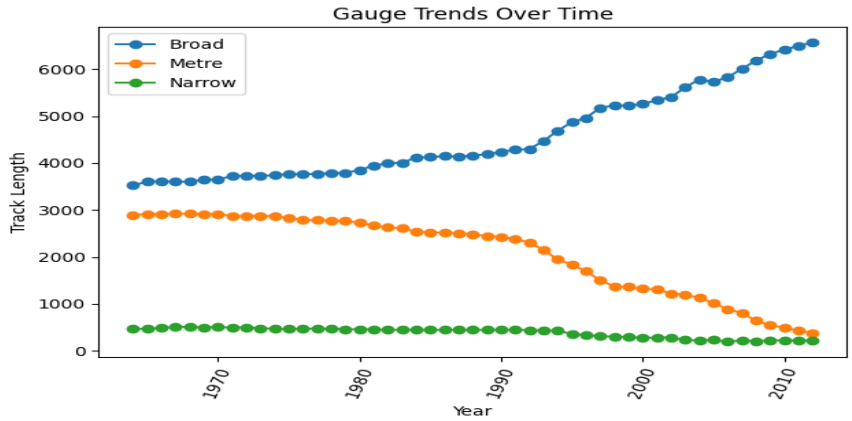
The pie chart clearly shows that Broad Gauge has the highest contribution among all gauge types.It occupies the largest portion of the chart, which means it dominates the railway network.Metre Gauge and Narrow Gauge have much smaller shares in comparison.

Scenario 5: Advanced Analysis + Multiple Graphs (● Hard)

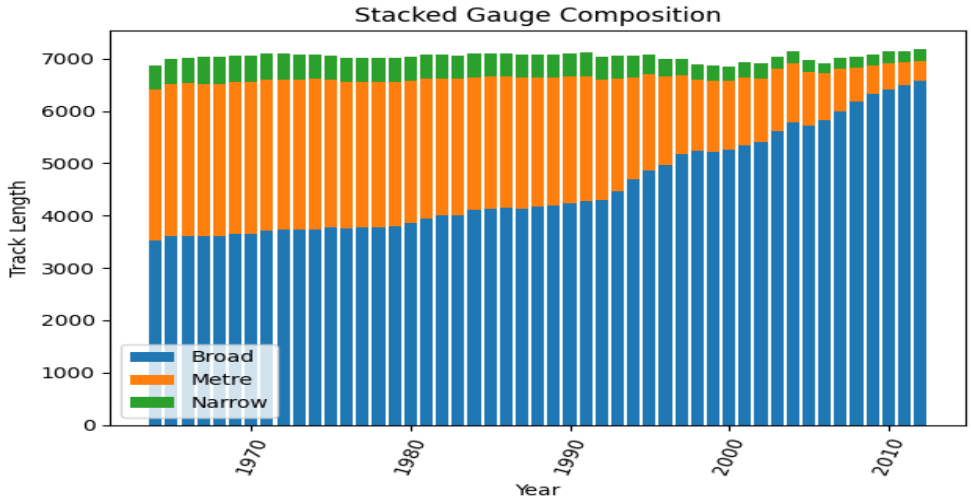
Description.

These percentages help us understand how much each gauge contributes to the total railway network.Using NumPy, yearly growth is calculated to analyze how railway tracks increase or decrease over time.A line graph shows trends of all gauges, while a stacked bar chart shows their yearly composition.

Task	Description
1.	**Create new columns** for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function np.diff to calculate the **yearly growth (change)** of the Total tracks column.
3.	Plot: a) A single **Line Graph** showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A **Stacked Bar Chart** showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	**Highlight/Identify:** a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	**Final Conclusion:** Based on the percentage composition line graph, provide a conclusion to the question: **"Is the railway system shifting towards a single ``` gauge?"**



Graph



Conclusion:

The analysis shows that Broad Gauge is steadily increasing and dominating the railway network.At the same time, Metre Gauge and Narrow Gauge are continuously decreasing.The graphs clearly indicate a shift in focus towards one uniform system. This change improves speed, capacity, and efficiency of rail transport.

What you have learned

Learned how to load and clean data using Pandas.Handled missing values and converted data into numeric for Used a line graph to understand the growth trend of railway tracks. Compared different gauges using bar charts and pie charts.Created new percentage columns and calculated yearly growth.Found that the railway system is gradually shifting towards Broad Gauge.